

SAT.O–Blockwave: Unified Action — Corrected, Dimensionally Consistent Formulation

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Abstract

A single, dimensionally consistent formulation of the SAT.O/Blockwave unified-action model. This source consolidates the full field content, specifies SI-consistent units for all quantities, presents the corrected Lagrangian density, and lists the resulting coupled partial differential equations. An appendix provides line-by-line dimensional bookkeeping.

Conventions and SI base units

We use SI base symbols for dimensional analysis: M (mass), L (length), T (time). The metric signature is not required for this non-relativistic morphogenic presentation.

1 Field definitions and identities

We define the basic fields used throughout:

$$\mathbf{v}(\mathbf{r}, t) = \frac{\mathbf{J}(\mathbf{r}, t)}{\rho(\mathbf{r}, t)}, \quad [\mathbf{v}] = LT^{-1}, \quad (1)$$

$$S[\Phi] \equiv (\nabla\Phi), \quad [\nabla\Phi] = LT^{-2}. \quad (2)$$

The curvature tensor for the scalar field is the rank-2 morphotensor $\mathbf{C} = \nabla \otimes \nabla\Phi$.

Symbol	Physical meaning and SI dimensions
ρ	mass (energy) density, $[\rho] = ML^{-3}$
$\mathbf{J}$	mass flux density, $[\mathbf{J}] = ML^{-2}T^{-1}$
$\mathbf{v}$	velocity field, $[\mathbf{v}] = LT^{-1}$
Φ	morphogenic potential per unit mass, $[\Phi] = L^2T^{-2}$
$W(\Phi)$	potential per unit mass (same units as Φ), $[W] = L^2T^{-2}$
$U(\Phi) = \rho W(\Phi)$	energy density, $[U] = ML^{-1}T^{-2}$
κ	curvature coupling, $[\kappa] = ML^{-3}T^2$
c	characteristic wave speed, $[c] = LT^{-1}$
γ	damping, $[\gamma] = T^{-1}$
λ	dimensionless coupling (appears in Lagrangian as $\lambda \rho \Phi \nabla \cdot \mathbf{v}$)
P	pressure, $[P] = ML^{-1}T^{-2}$

Table 1: Primary symbols and their SI-base dimensions used in the text.

2 Corrected Lagrangian density

We choose a dimensionally consistent Lagrangian density (energy density units):

$$\mathcal{L} = \frac{1}{2}\rho\mathbf{v}^2 - \rho W(\Phi) - \frac{\kappa}{2}(\nabla\Phi)^2 + \lambda\rho\Phi(\nabla\cdot\mathbf{v}). \quad (3)$$

Here $W(\Phi)$ is the potential per unit mass; the coupling λ is taken dimensionless and ρ supplies appropriate units in the interaction term.

3 Euler–Lagrange system (resulting PDEs)

Variation of $\mathcal{L}$ with respect to the hydrodynamic variables and Φ yields the following coupled PDE system.

Mass continuity

$$\partial_t\rho + \nabla\cdot(\rho\mathbf{v}) = 0. \quad (4)$$

Momentum balance

The momentum equation (obtained from $\delta\mathcal{L}/\delta\mathbf{v}$) reads:

$$\rho(\partial_t\mathbf{v} + (\mathbf{v}\cdot\nabla)\mathbf{v}) = -\nabla P - \rho\nabla W(\Phi) - \kappa(\nabla^2\Phi)\nabla\Phi + \lambda\nabla(\rho\Phi), \quad (5)$$

where the curvature-derived body force has the form $\mathbf{f}_\kappa = -\kappa(\nabla^2\Phi) \nabla\Phi$ (see Appendix for derivation). The last term arises from the interaction piece and is dimensionally consistent when λ is dimensionless.

Scalar (per-unit-mass) evolution

The scalar evolution equation (variation w.r.t. Φ) in mass-normalized form is:

$$\partial_t^2\Phi + \gamma \partial_t\Phi = c^2 \nabla^2\Phi - \frac{dW}{d\Phi} + \frac{\lambda}{\rho} \nabla \cdot (\rho\mathbf{v}). \quad (6)$$

Here $dW/d\Phi$ carries units of T^{-2} so that each term has the same dimensions $[\Phi]/T^2 = L^2T^{-4}$.

4 Energy and conserved quantity

Define total energy (for $\gamma = 0$) as

$$E_{\text{tot}} = \int_V \left[\frac{1}{2}\rho\mathbf{v}^2 + \rho W(\Phi) + \frac{\kappa}{2}(\nabla\Phi)^2 \right] dV. \quad (7)$$

Direct computation, using the PDEs above, shows that $dE_{\text{tot}}/dt = 0$ when $\gamma = 0$ and appropriate boundary conditions apply (see Appendix).

Dimensionless rescaling

Introduce characteristic scales L_0 , T_0 , ρ_0 , and Φ_0 . Define nondimensional variables

$$\tilde{\mathbf{r}} = \frac{\mathbf{r}}{L_0}, \quad \tilde{t} = \frac{t}{T_0}, \quad \tilde{\rho} = \frac{\rho}{\rho_0}, \quad \tilde{\Phi} = \frac{\Phi}{\Phi_0}, \quad \tilde{\mathbf{v}} = \frac{\mathbf{v}T_0}{L_0}.$$

The governing equations can be written in terms of dimensionless control numbers such as

$$\text{Re}_\Phi = \frac{\rho_0 L_0^2}{\kappa T_0^2}, \quad \text{Ma}_\Phi = \frac{V_0}{c}, \quad \text{Da}_\Phi = \frac{W(\Phi_0)T_0^2}{\Phi_0}.$$

Appendix: Dimensional bookkeeping and derivations

A. Units for the curvature force

The curvature energy density is $\frac{\kappa}{2}(\nabla\Phi)^2$. Variation with respect to Φ yields a functional derivative that produces a body-force density of the form

$$\mathbf{f}_\kappa = -\kappa (\nabla^2\Phi) \nabla\Phi + (\text{higher-order gradient terms}).$$

Check units: $[\nabla^2\Phi] = T^{-2}$, $[\nabla\Phi] = LT^{-2}$, $[\kappa] = ML^{-3}T^2$ so $[\mathbf{f}_\kappa] = ML^{-2}T^{-2}$ as required for a force density.

B. Euler–Lagrange to PDE mapping

A short derivation of the scalar equation: compute $\delta\mathcal{L}/\delta\Phi$ and apply continuity where necessary; one finds the mass-normalized wave-like equation shown in Eq. (6).

C. Energy conservation sketch

Multiply Eq. (5) by $\mathbf{v}$ and integrate, combine with Eq. (6) multiplied by $\rho \partial_t\Phi$, and integrate by parts. Under vanishing-flux boundary conditions the total energy time derivative vanishes for $\gamma = 0$.